$$R_{\text{core}} = \frac{G M m_0}{E_H}$$

$$\rho_{\text{core}} = \frac{M}{R_{\text{core}}^3}$$

$$p_{\text{core}} = \frac{1}{3}\,\rho_{\text{core}}\,c^2 = \frac{1}{3}\,\frac{M}{R_{\text{core}}^3}\,c^2$$

Parametr:

$$\eta = \frac{p_{\text{core}}\,R_{\text{core}}^4}{G M^2}$$

Podstawiamy:

$$\eta = \frac{\frac{1}{3}\frac{M}{R_{\text{core}}^3}\,c^2 \cdot R_{\text{core}}^4}{G M^2} = \frac{1}{3}\,\frac{M c^2 R_{\text{core}}}{G M^2} = \frac{1}{3}\,\frac{c^2 R_{\text{core}}}{G M}$$

I dalej:

$$R_{\text{core}} = \frac{G M m_0}{E_H} \;\Rightarrow\; \eta = \frac{1}{3}\,\frac{c^2}{G M} \cdot \frac{G M m_0}{E_H} = \frac{1}{3}\,\frac{c^2 m_0}{E_H}$$

Extreme energy density and temperature during gravitational collapse trigger a reverse Higgs phase transition (symmetry restored to v = 0). The resulting sudden loss of particle rest mass and the emergence of massless relativistic radiation pressure halt further collapse and prevent singularity formation.

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